Algebra: Piecewise Defined Functions

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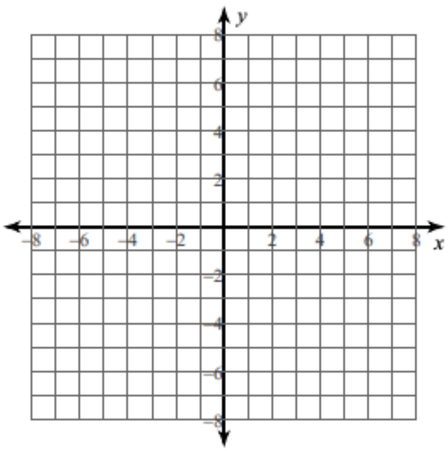
Name: ______________________________ Date: __________________ Score: / 8

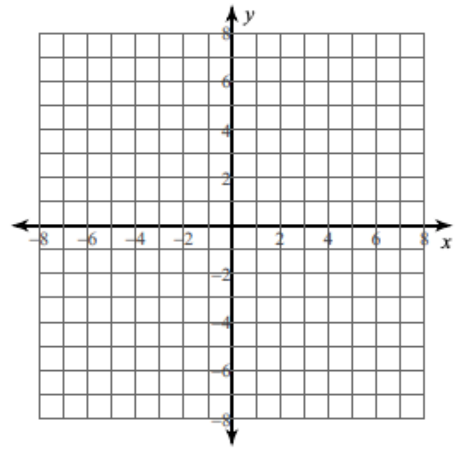
DIRECTIONS *Evaluate the piecewise function at each x-value by choosing the correct piece.*

Sketch the graph of each function.

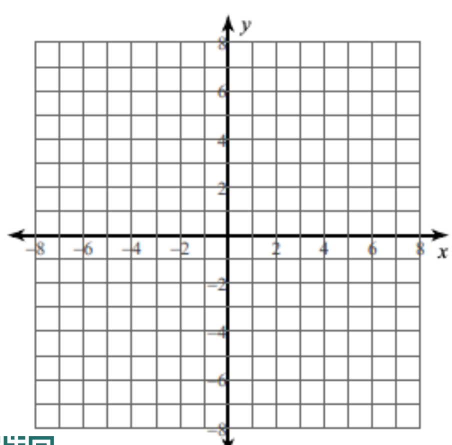
1) $f(x) = \begin{cases} -2x - 1, & x \le 2 \\ -x + 4, & x > 2 \end{cases}$ f (3) = $\boxed{}$



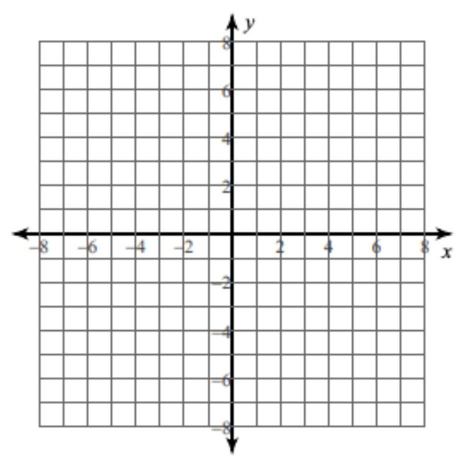
2) $f(x) = \begin{cases} -4, & x \le -2 \\ x - 2, & -2 < x < 2 \\ -2x + 4, & x \ge 2 \end{cases}$ f (2) = $\boxed{}$



3) $f(x) = \begin{cases} -2^x, & x < -4 \\ -|x|, & -4 \le x \le 0 \\ 4 - x^2, & x > 0 \end{cases}$ f (-2) = $\boxed{}$



4) $g(x) = \begin{cases} -6, & x < -2 \\ (x + 1)^4, & x \ge -2 \end{cases}$ f (-2) = $\boxed{}$



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Answer Key & Solutions

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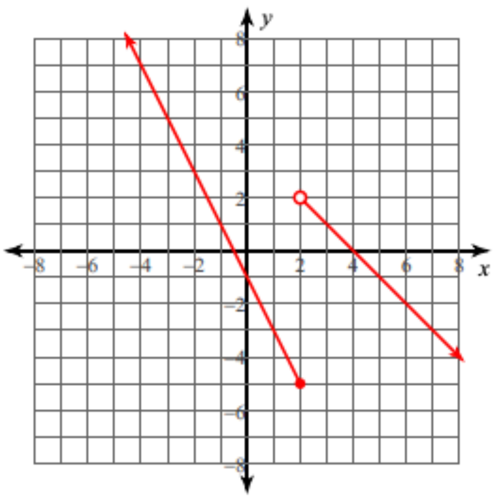
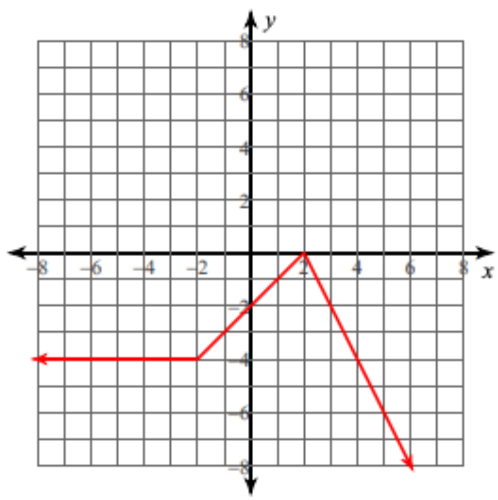
TEACHER NOTES *Students must check the condition for x first, then substitute into the matching rule.*

Sketch the graph of each function.

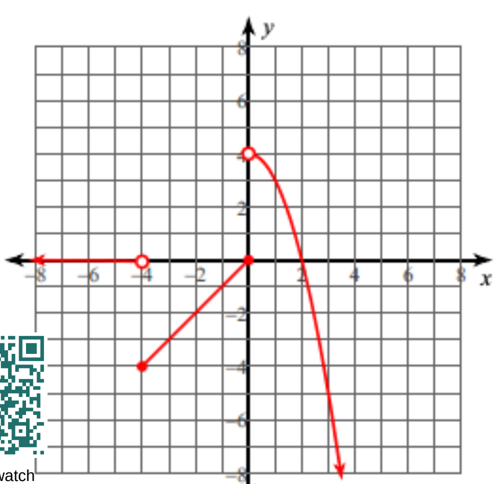
1) $f(x) = \begin{cases} -2x - 1, & x \le 2 \\ -x + 4, & x > 2 \end{cases}$ f (3) = $\boxed{1}$



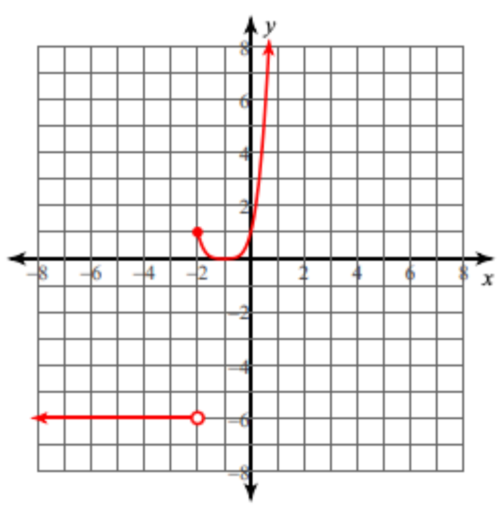
2) $f(x) = \begin{cases} -4, & x \le -2 \\ x - 2, & -2 < x < 2 \\ -2x + 4, & x \ge 2 \end{cases}$ f (2) = $\boxed{0}$



3) $f(x) = \begin{cases} -2^x, & x < -4 \\ -\lvert x \rvert, & -4 \le x \le 0 \\ 4 - x^2, & x > 0 \end{cases}$ f (-2) = $\boxed{-2}$



4) $g(x) = \begin{cases} -6, & x < -2 \\ (x + 1)^4, & x \ge -2 \end{cases}$ f (-2) = $\boxed{1}$



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